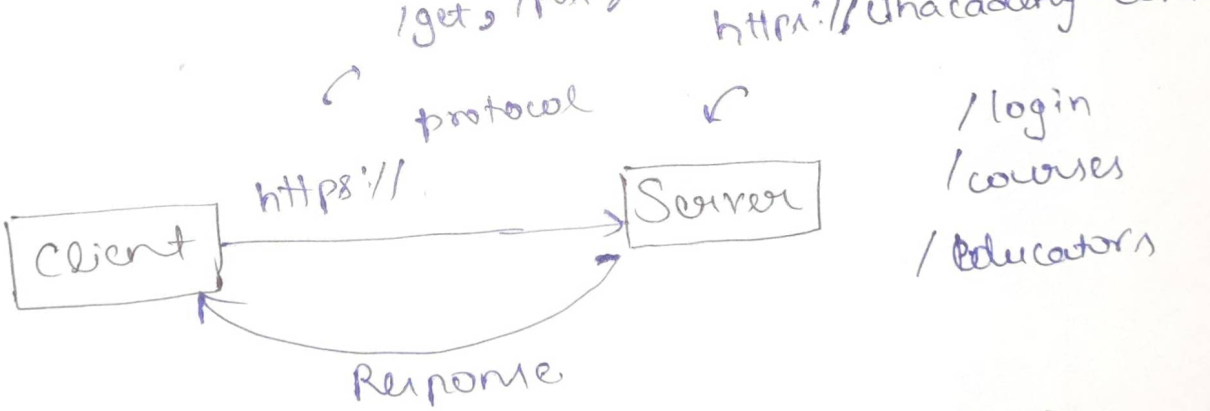
MCP (Model Context Protocol)

→ MCP is an open-source standard for connecting AI applications to external systems.

→ Using MCP, AI applications like Claude or ChatGPT can connect to data source (e.g. local files, DB), tools (e.g. search engines) & workflows (e.g. specialized prompts) - enabling them to access key info. and perform tasks.

→ Think of MCP like USB-C port for AI app^n. Just as USB-C provides a standardized way to connect electronic devices, MCP provides a standardized way to connect AI app^n to external systems.

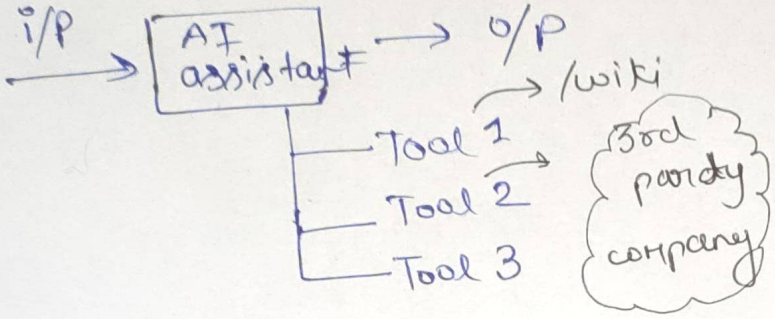
⇒ Idea behind protocol:



we will get response in form of JSON (using REST APIs)

⊨ Gen AI & Agentic AI applications: (happening ?)

Generative AI

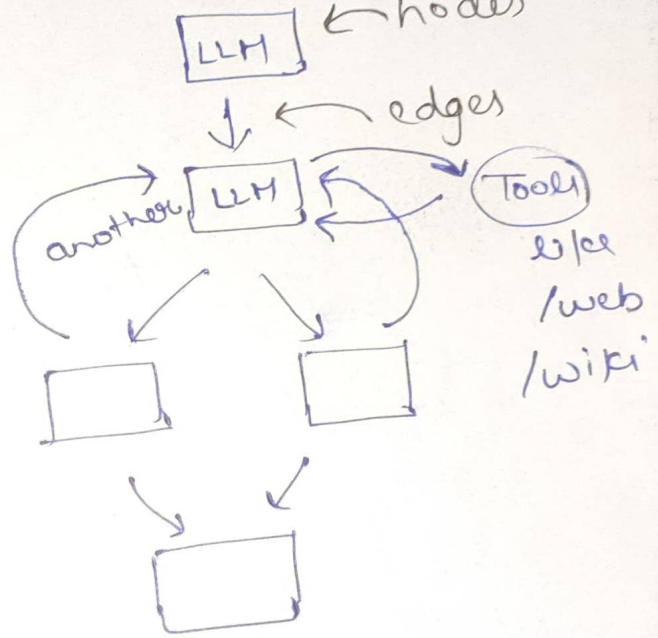


- whenever 3rd party company like arxiv, wiki, tavily, vectorDB ⇒ give their response — we have to update the code.

Agentic AI

→ Stateful AI agents (1st day using LangGraph)

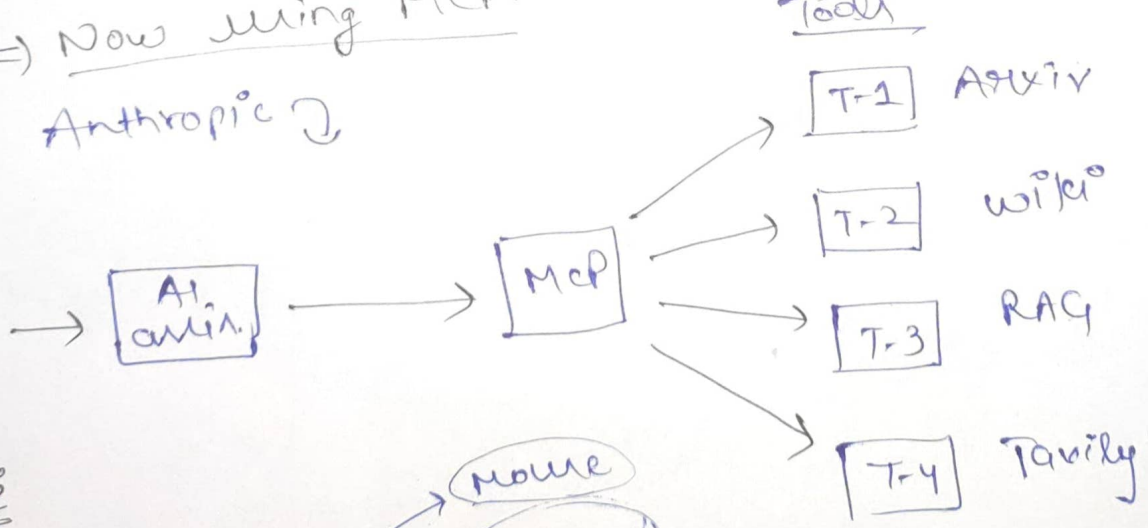
→ solving complex workflow



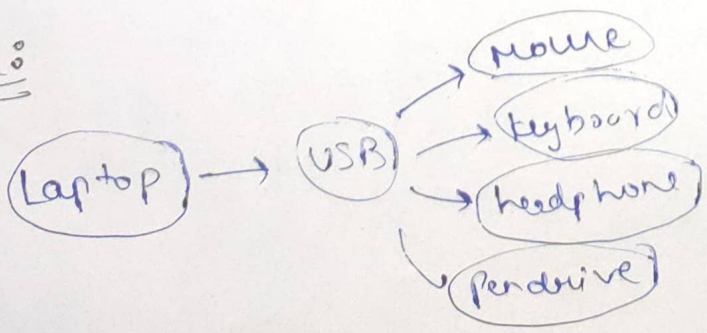
→ every node behave like an AI agent

- ⊨ Now using MCP:

Anthropic ↴



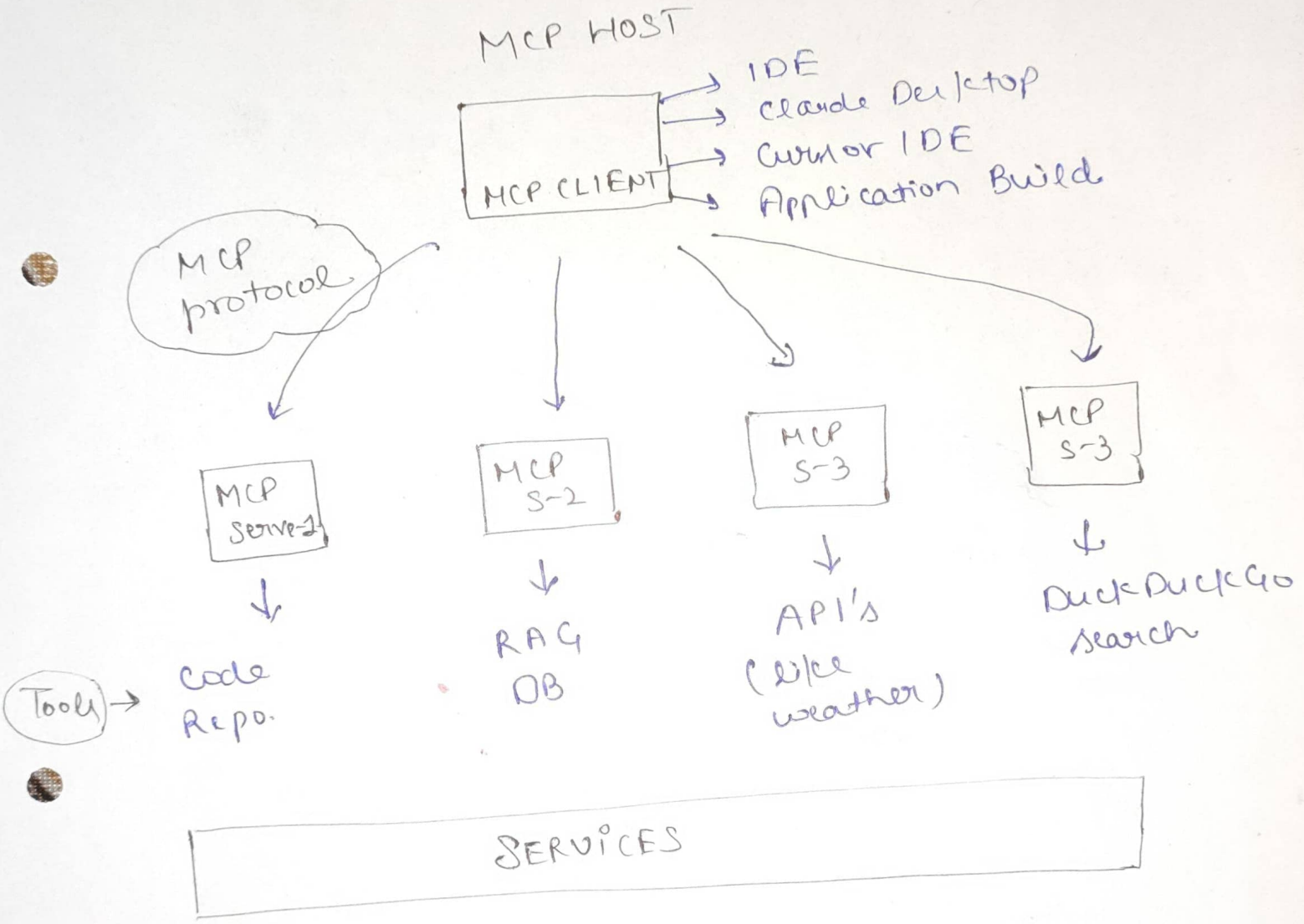
ex:



these tool managed by the {service providers} (so we do not need to make any changes in client end)

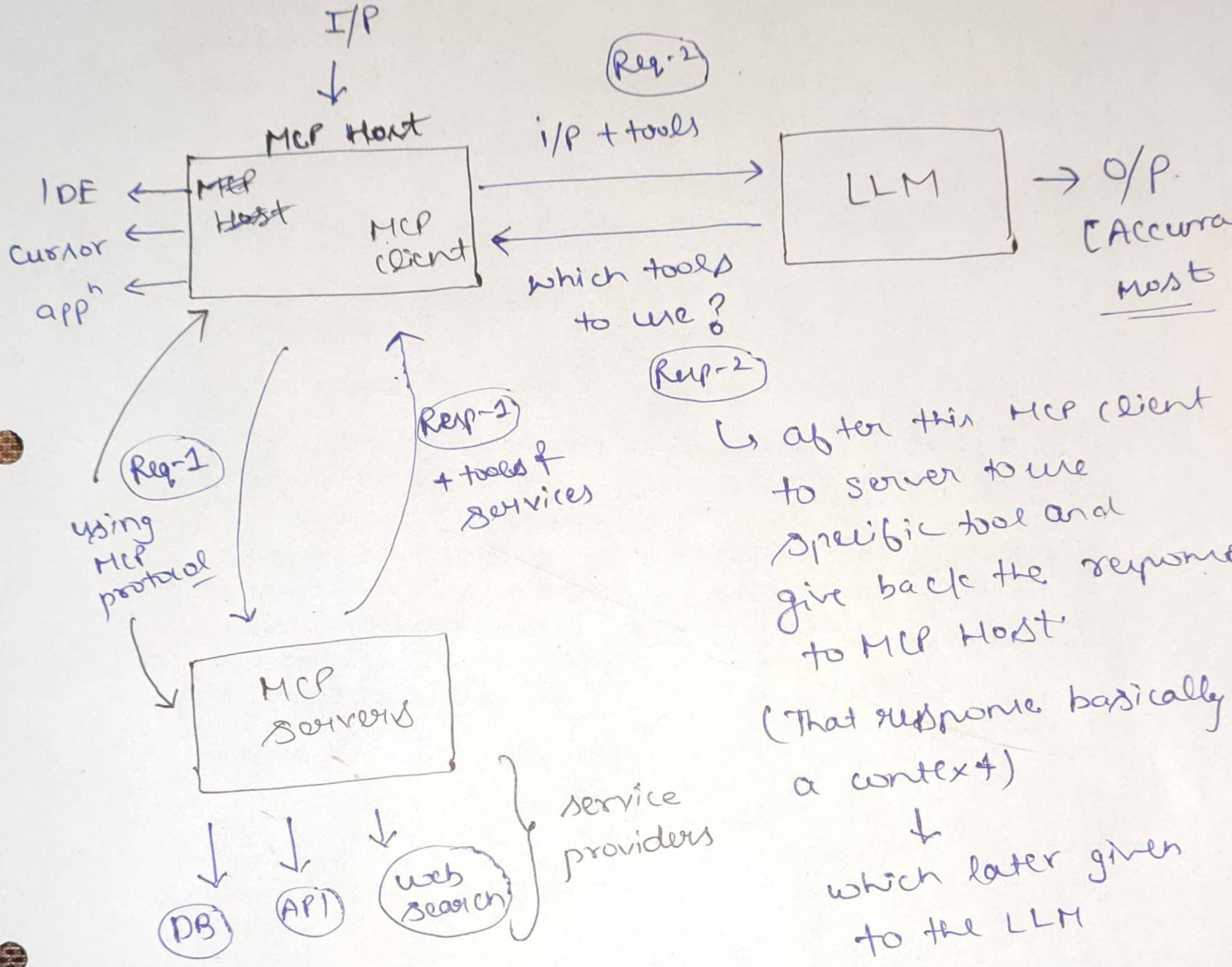
Important Components of MCP :-

i) MCP client
ii) MCP Host → It can be IDE / Claude Desktop / Cursor IDE / any app^n
iii) MCP Server



→ above Tools & Services are managed by Service Providers

Communication B/w the components:-



↳ after this MCP client to server to use specific tool and give back the response to MCP Host

(That response basically a context)

↓

which later given to the LLM

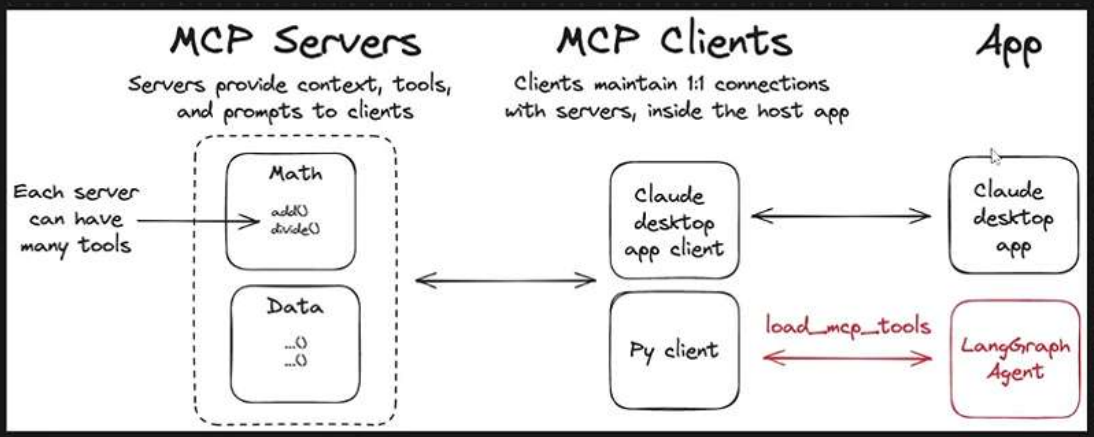
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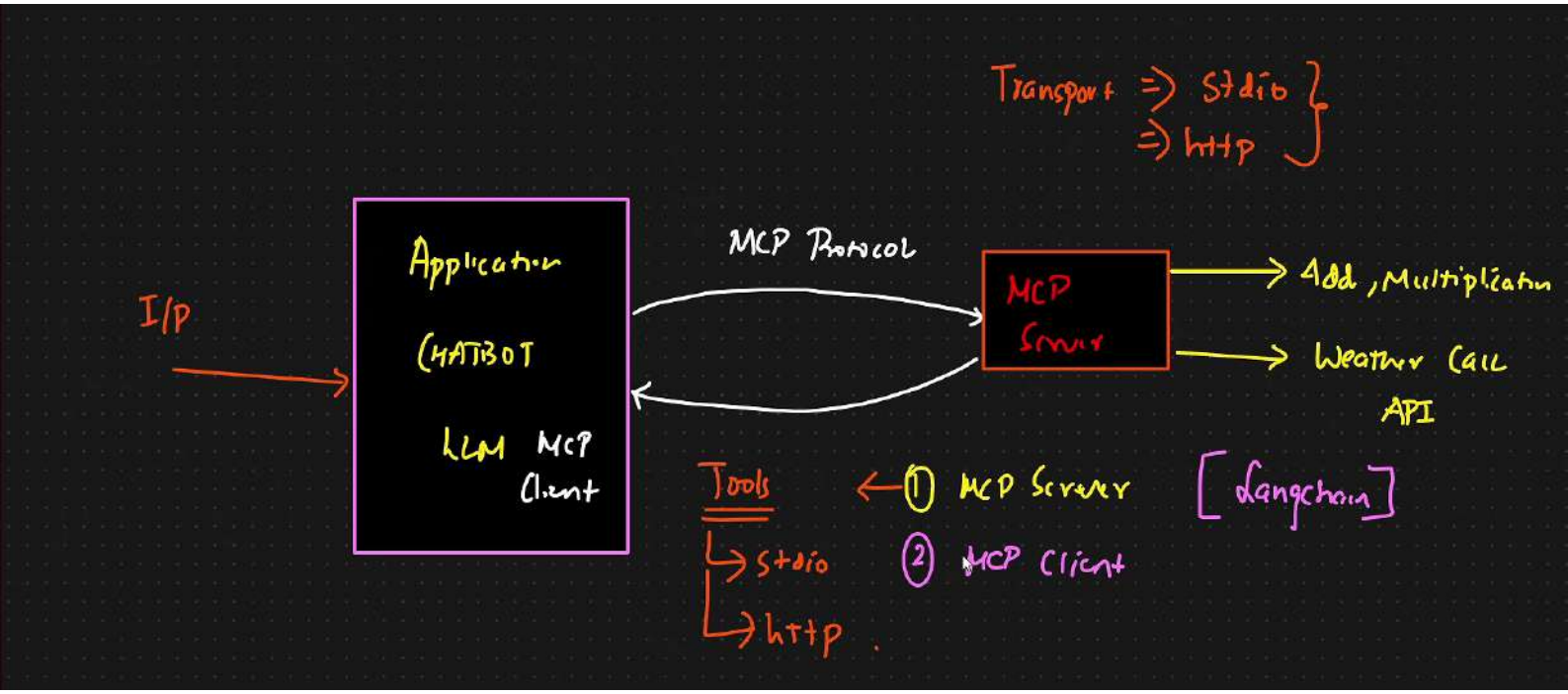
O/P

(most accurate O/P)

→ MCP client is used to interacting with the MCP servers with the help of MCP protocol

Building MCP Server [MCP Integration]




Transport ⇒ Stdio
⇒ http
I/P
Application
CHATBOT
LLM MCP Client
MCP Protocol
MCP Server
Add, Multiplication
Weather Call API
Tools
stdio
http
① MCP Server
② MCP Client
[langchain]